

DENOISING TOMOGRAPHY TO ENHANCE INSIGHTS

Brains on Silicon 2025

- ❑ ULTRAFAST CT (ROFEX) AT HZDR
- ❑ USING MACHINE LEARNING TO IMPROVE NOISY DATA
- ❑ IMPACT ON SCIENCE & INDUSTRY APPLICATIONS

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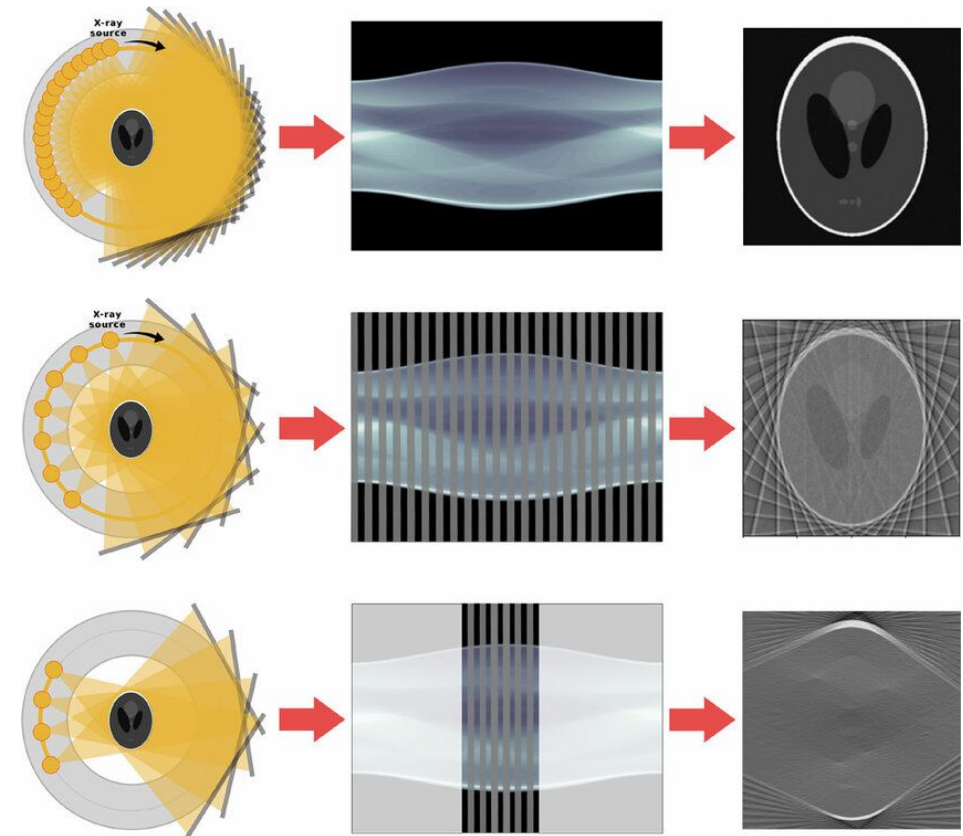
TOMOGRAPHY

TOMOGRAPHIC IMAGE RECONSTRUCTION WORKFLOW

- An imaging technique that creates cross-sectional (2D) views of an object.
- It reconstructs 2D slice from multiple 1D projections taken at different angles.
- The combined 1D projections form a sinogram.
- The reconstructed 2D image from the sinogram is called a tomogram.

Basic Principle:

- The object is scanned from different angles.
- Computer algorithms reconstruct the internal structure.
- **Examples:** CT (X-ray), PET (gamma rays), MRI (magnetic fields)



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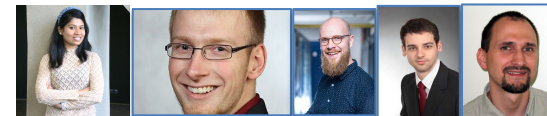
ML-BASED DENOISING OF TOMOGRAPHY DATA

ULTRAFAST ELECTRON BEAM X-RAY COMPUTED TOMOGRAPHY (ROFEX) AT HZDR

Why it matters:

- Tomography data is often noisy
- Noise limits image quality & reconstruction accuracy
- Impacts research & industry (healthcare, material science, semiconductors)

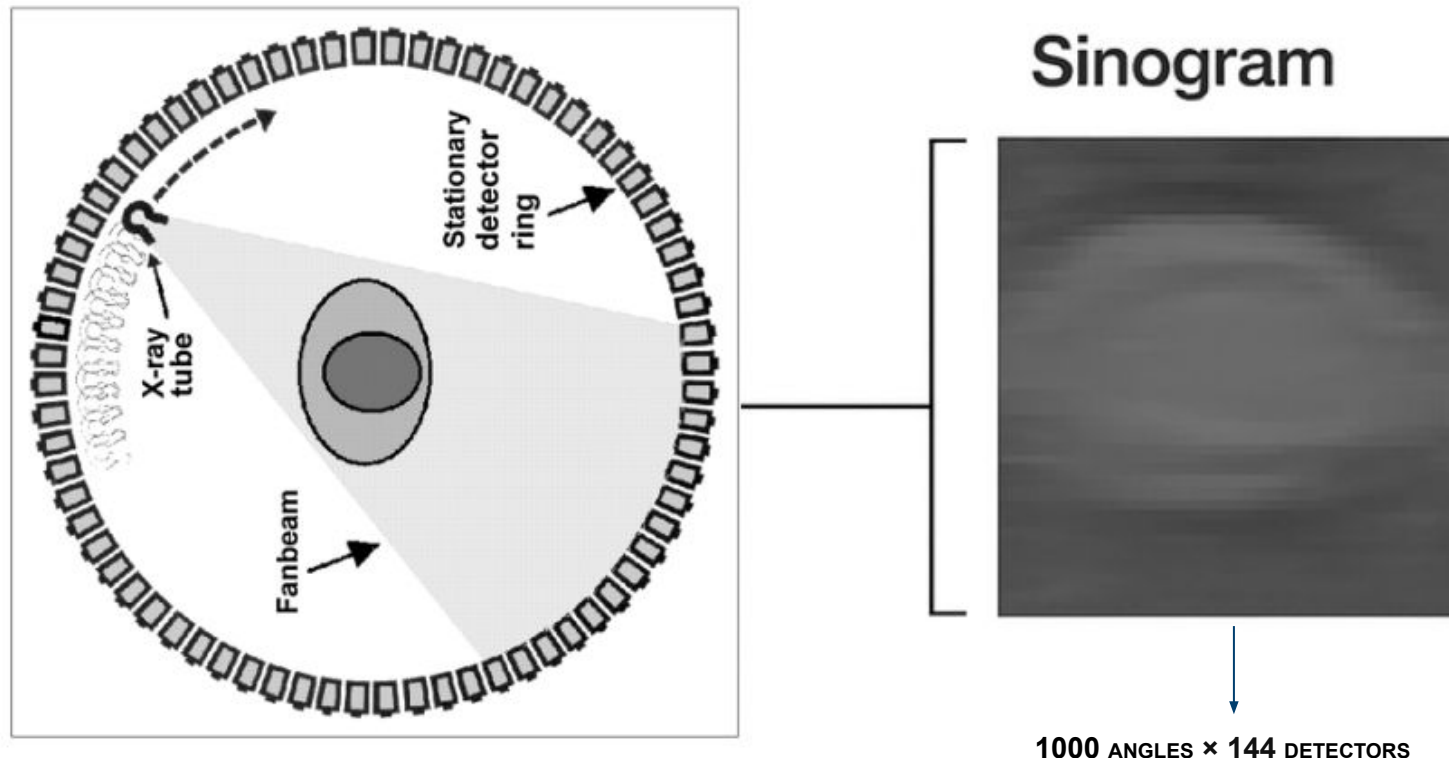
Our aim: Clean the sinogram first before reconstruction.



SINOGRAM

HOW A SINOGRAM IS MADE

- One 1-D projection angle: the detector gives **144** values (one per bin).
- Stack 1000 angles → one sinogram: size **1000 × 144** (angles × detectors).
- Repeat 2000 times → full dataset: **(2000, 1000, 144)** = (sinograms, angles, detectors).



METHODS

CLASSICAL VS ML (SELF-SUPERVISED)

- **Baselines: BM3D**¹ / simple temporal or angular averaging.
- **Our approach:** self-supervised **Noise2Void**^{2,3} / **N2V2** (CAREamics⁴) on the sinogram before reconstruction → learns from noisy data, no clean target needed.
- **Where applied?** On sinograms (pre-reconstruction) to make the approach independent of the reconstruction algorithm.

¹https://github.com/ahendriksen/pybind_bm3d

²Krull, A., Buchholz, T.-O., & Jug, F. (2019). Noise2Void: Learning denoising from single noisy images. In CVPR 2019. Link

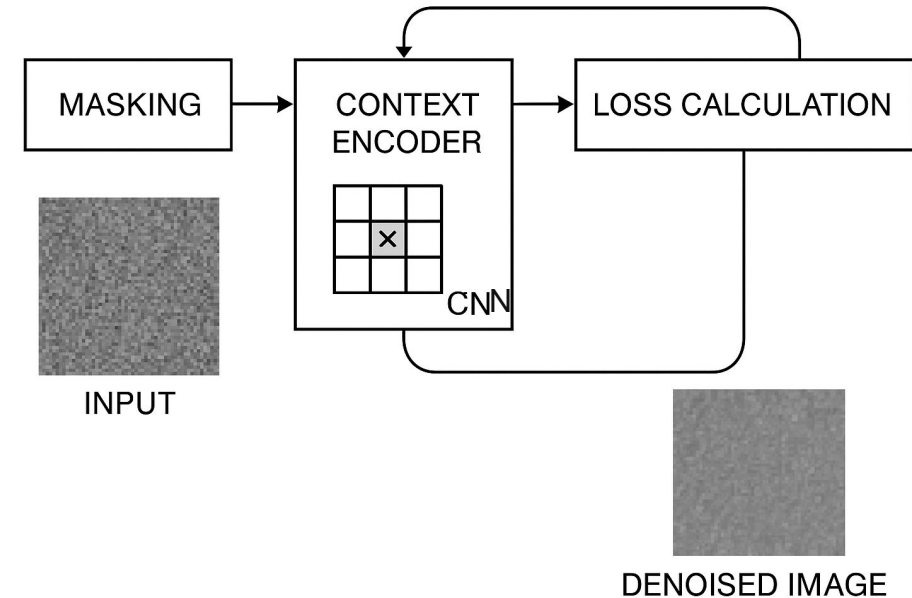
³Batson, J., & Royer, L. (2019). Noise2Self: Blind denoising by self-supervision. In ICML 2019, PMLR 97:524–533. Link

⁴<https://careamics.github.io/0.1/algorithms/>

WHY SELF-SUPERVISED?

NO CLEAN TARGETS NEEDED

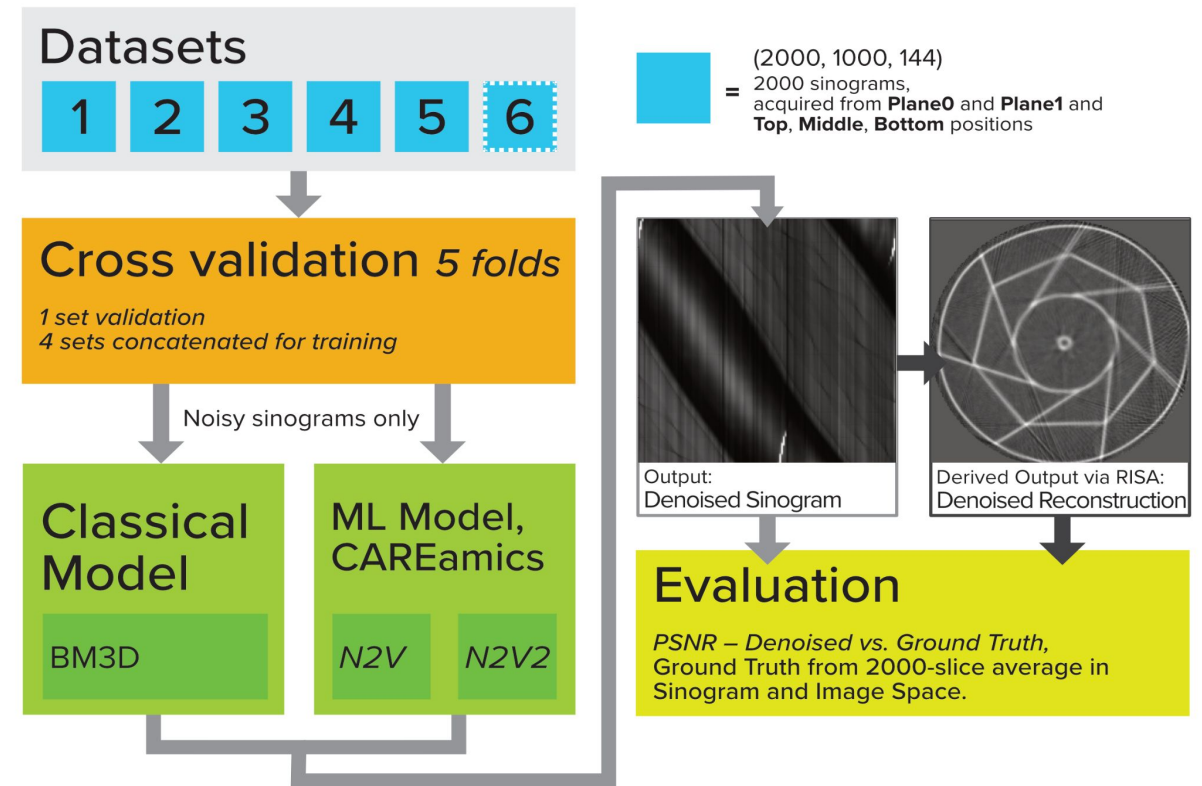
- A few pixels are hidden in a noisy patch so the network can't copy them.
- The network predicts the hidden pixels from their neighbors; loss is computed only on those masked spots.
- Because noise is random, matching many masked pixels teaches the network the underlying clean structure—so we don't need a clean ground-truth image.



How Our Denoising Workflow Works

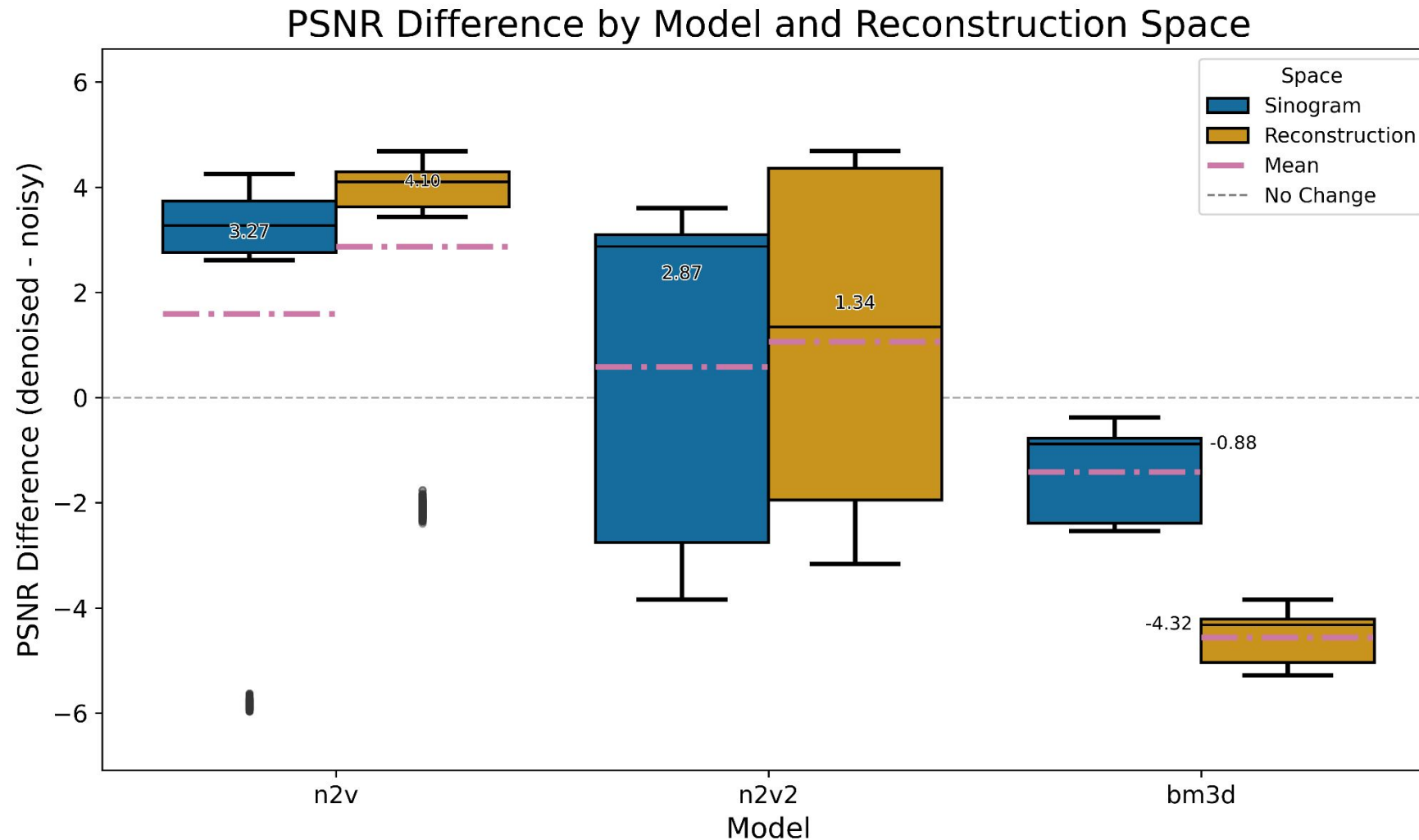
STEP-BY-STEP PROCESS FROM SINOGRAMS TO CLEAN RECONSTRUCTION

- Dataset split: 6 folds → 1 for testing, remaining 5 for training + validation
 - Trained deep learning models directly on noisy sinogram data
 - Created a proxy clean reference by averaging 2000 sinograms (noise cancels, structure remains)
 - Compared against non-learning baseline (BM3D)
- **Reconstruction of denoised sinograms led to improved image quality**



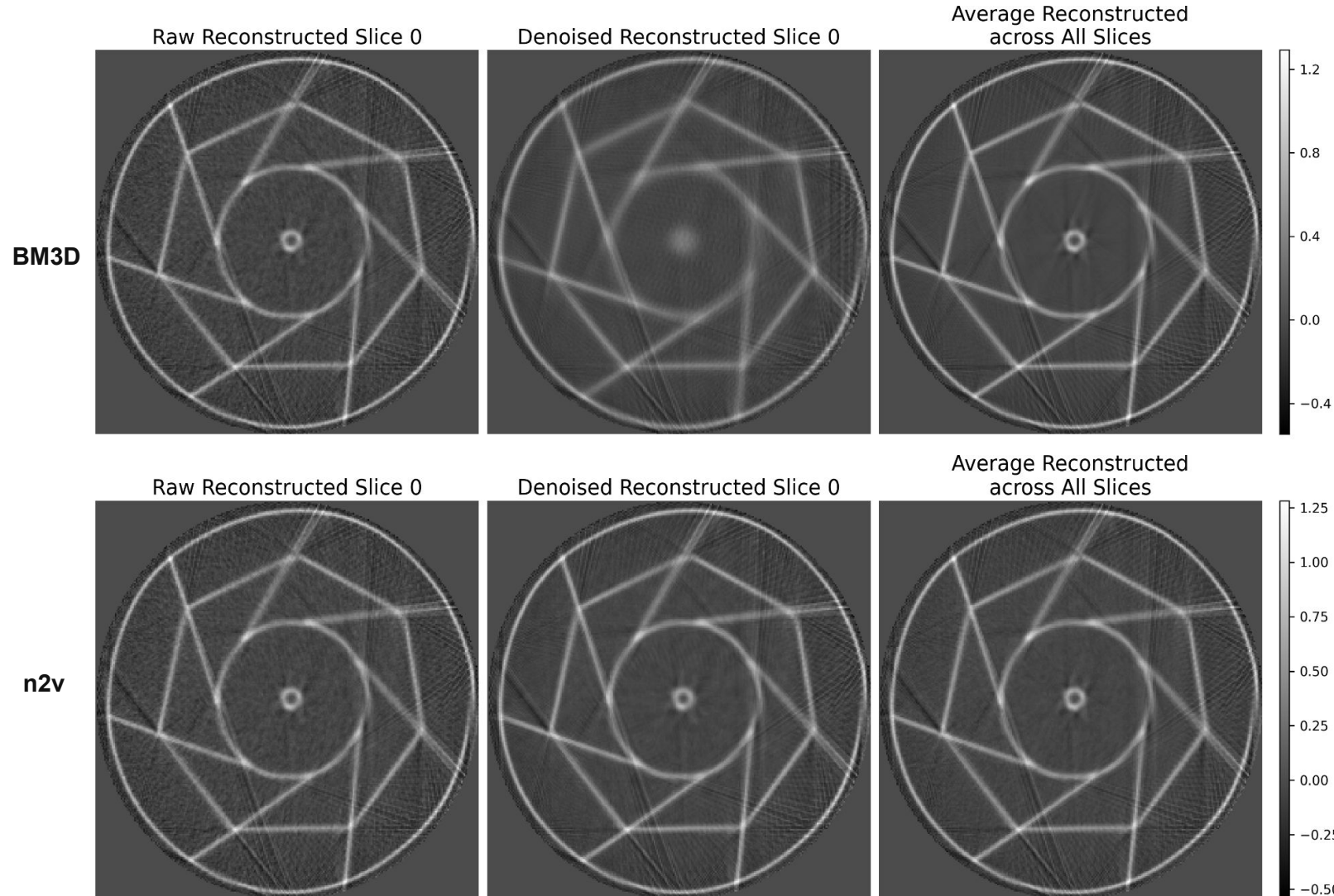
PERFORMANCE EVALUATION: How Much Better Is Deep Learning?

PSNR DIFFERENCE SHOWS THE IMPROVEMENT IN IMAGE QUALITY



VISUAL COMPARISON: DEEP LEARNING VS. CLASSICAL DENOISING

HOW DIFFERENT METHODS AFFECT IMAGE QUALITY



- Deep learning outperforms traditional non-learning methods
- Higher PSNR → Clearer Images
- Planned applicability to industrial workflows (microscopy, metrology, semiconductors)

Future Step: Working on deep learning architectures for better generalization

Thank You!

With special thanks to my collaborators

Empowering industry
collaboration through HPC
and AI – unlocking new
possibilities in tomography.



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